CostPilot — Phase 2 Plan

From Analysis Copilot to Cost & Performance Platform

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- 1. Executive Summary
- 2. What Phase 2 Adds, and Why
- 3. Scope
  - 3.1 Module 4 — Activity-Based Costing (the headline feature)
  - 3.2 Module 5 — Budgeting Workflows & Rolling Forecast
  - 3.3 Module 6 — Performance Measurement
  - 3.4 Module 7 — ERP & Data Connectors
  - 3.5 Module 8 — Continuous Monitoring & Alerts
  - 3.6 Module 9 — Group-Level Reporting Aids
- 4. Non-Functional Goals for Phase 2
- 5. Delivery Plan
- 6. Success Criteria
- 7. Key Risks and Mitigations
- 8. Looking Beyond: Phase 3 Preview

1. Executive Summary

Phase 1 proves the concept: explainable variance analysis and short-run decision support on tabular cost data. Phase 2 turns the proof into an enterprise-useful product by attacking the pains that require deeper data and broader process coverage — distorted product profitability (activity-based costing), slow budgeting cycles, performance measurement that games managers, and the manual ERP-to-analysis pipeline.

Phase 2 keeps every Phase 1 design rule intact: the deterministic engine owns all arithmetic, the LLM only plans and narrates, and no answer ships without its audit trail.

Phase 2 outcome: a copilot a cost-accounting team can run a real close, a real reforecast, and a real product-pricing review on — connected to their actual systems.

2. What Phase 2 Adds, and Why

#	Pain carried over from Phase 1	Phase 2 capability
1	Overhead allocation distorts product profitability — the Siemens problem: custom/low-volume products under-costed by multiples while standard products are over-costed and priced out of the market	ABC / TDABC product & customer profitability
2	Budget and reforecast cycles take weeks; consolidation of hundreds of spreadsheet versions is manual; forecasts go stale	Budgeting workflows + rolling forecast

#	Pain carried over from Phase 1	Phase 2 capability
3	ROI-based divisional evaluation causes managers to reject profitable investments; Balanced Scorecards die as manual reporting rituals	Performance-measurement module (ROI vs Residual Income, lightweight BSC)
4	Data entry via CSV is acceptable for a pilot, unacceptable at production scale	ERP connectors (SAP, Oracle, NetSuite) with governed data contracts
5	Management P&L versus statutory P&L reconciliation remains partly manual	Automated costing-method bridge at group level
6	Variances and anomalies are found only at period end	Continuous monitoring with alerting

3. Scope

3.1 Module 4 — Activity-Based Costing (the headline feature)

Full ABC pipeline as a guided, conversational modeling workflow:

1. Identify activities from ledger + operational data (setups, inspections, purchase orders, order processing, engineering support).
2. Build cost pools per activity.
3. Select cost drivers per pool (number of setups, inspections, purchase orders, customer orders, material movements).
4. Compute activity cost-driver rates: total activity cost ÷ total driver units.
5. Allocate to products/customers by actual driver consumption.
6. Output: product-profitability report comparing traditional-volume allocation versus ABC, exposing overcosted and undercosted lines — the exact Siemens EMW analysis.

TDABC as the default mode. Full ABC historically fails in practice because driver data collection is too expensive to maintain. Time-Driven ABC estimates only two inputs per process step (capacity cost rate and time per transaction), which makes the model maintainable. The copilot's role: elicit process times conversationally, flag unrealistic estimates, and keep the model current as transaction volumes change.

Customer profitability analysis using the same machinery — orders, service tickets, delivery frequency as drivers — extending product profitability to account-level margin truth.

Prerequisite work: driver-event capture from operational systems (setup logs, inspection counts, order lines), which overlaps with Module 7 (ERP connectors) and should be sequenced with it.

3.2 Module 5 — Budgeting Workflows & Rolling Forecast

- Budget template system: sales budget → production budget (Sales + Desired Closing – Opening inventory) → materials purchase budget → labour budget → cash budget → master budget, each computed by the engine with the standard formulas, reviewed and adjusted in-conversation.
- Multi-sheet consolidation with version control: no more conflicting workbook copies; the copilot answers "what changed between Budget v3 and v4 and who changed it".
- Participative-budget games detection: compare submitted targets against historical run-rates and flag probable sandbagging for reviewer attention (flag, not decide).
- Rolling forecasts: on each period close, the engine recomputes remaining-period projections from actuals plus user-updated assumptions; what-if scenarios (price, volume, input-cost shocks) run in

seconds against the same deterministic core.
- Budget-versus-actual monitoring becomes continuous (see Module 8), not a period-end exercise.

3.3 Module 6 — Performance Measurement

- **Divisional evaluation suite:** ROI, Residual Income, and profit-margin comparisons on the same data, with the copilot explicitly demonstrating where ROI maximization rejects investments that Residual Income would accept — the decision-quality argument made numerically per case rather than taught as theory.
- **Balanced Scorecard, kept lightweight:** the Phase 1 lesson applies — BSC initiatives die when they create reporting work. So the copilot auto-pulls financial KPIs from the data layer, customer and process KPIs from connector feeds, and asks humans only for what cannot be measured automatically. Strategy-map cause-and-effect chains rendered from user-declared objectives.
- Weighted scorecard computation with user-set weights, transparent per-metric contribution to the overall score.
- Controllability tagging: costs tagged controllable/uncontrollable per manager so variance accountability follows the Caterpillar principle — evaluate managers on controllable execution, never on volume swings.

3.4 Module 7 — ERP & Data Connectors

- Priority order from pilot demand: SAP (large manufacturers), Oracle Fusion/NetSuite, then generic ODBC/JDBC and flat-file-on-schedule fallback.
- Governed data contracts per connector: chart-of-accounts mapping into the Phase 1 registry (fixed/variable, direct/indirect, dimensions) with mapping suggestions by the LLM and confirmation by a human — mapping decisions are config, never inferred silently.
- Incremental extraction with reconciliation checks against ERP control totals; every ingestion run produces a data-quality report.
- Read-only by design in Phase 2. No copilot write-back to ledgers.

3.5 Module 8 — Continuous Monitoring & Alerts

- Variance and anomaly detection on a rolling basis (post-ingestion, not post-close): standard-cost deviations beyond thresholds, spend against budget burn-rate, margin drift by product.
- Management-by-exception routing: alerts summarised, ranked by materiality, and delivered with the same differential-analysis explanation already built in Phase 1 — the alert *is* a mini variance pack.
- Natural-language alert rules: “tell me weekly if any product’s contribution margin drops more than 3 points” compiled into engine queries.

3.6 Module 9 — Group-Level Reporting Aids

- Automated variable-to-absorption profit bridges at entity and group level, consolidating the Phase 1 single-entity bridge.
- Multi-entity roll-ups with intercompany eliminations surfaced (not fully automated — flagged for review in Phase 2, automated in Phase 3).
- Standard management-report pack generation (monthly P&L by product/segment/plant with variance commentary) as an exportable document with live source links.

4. Non-Functional Goals for Phase 2

Area	Target
Security & access	Role-based views (a plant manager sees their plant); connectors use least-privilege read accounts; audit log of every query and generated answer
Data governance	Account-registry and driver changes are versioned, approved, and traceable
Performance	ERP-scale ingestion (millions of GL lines) remains within the Phase 1 answer-speed targets for pre-aggregated queries; heavy scans run in scheduled marts
Reliability	Same-question-same-number guarantee preserved across connector data refreshes via period snapshotting

5. Delivery Plan

Milestone	Contents	Exit criteria
M6 — Connectors foundation	SAP or NetSuite read connector, account-mapping workflow, ingestion quality reports	A live ERP extract loads into the registry and reconciles to ERP control totals
M7 — ABC/TDABC engine	Cost pools, driver rates, allocation, traditional-vs-ABC comparison report; TDABC capture flow	Reproduces the Siemens-style finding on pilot data: previously hidden undercosting of low-volume products quantified
M8 — Continuous variance & alerts	Rolling detection, materiality-ranked alert digest, NL alert rules	Pilot cost team receives a weekly exception digest with no manual report run
M9 — Budgeting & rolling forecast	Template chain, versioned consolidation, scenario reforecast	One full reforecast cycle completed in days instead of weeks with pilot team
M10 — Performance module	ROI/RI comparison, controllability tagging, auto-pulled scorecard	Divisional review meeting run on copilot output at pilot
M11 — Group reporting + hardening	Multi-entity bridges, report-pack export, RBAC, governance polish	Phase 1 + Phase 2 pilot team operating weekly without CSV fallback

Sequencing rationale: connectors (M6) come first because ABC driver data, alerts, and scorecard KPIs all depend on richer ingestion. ABC is kept as the headline deliverable (M7) because it is the most demonstrable Phase 2 value — the “400% under-costed product” reveal is this product’s strongest sales story.

6. Success Criteria

- **Correctness:** golden-case tests extended to ABC allocations, budget chain, ROI/RI; engine remains the sole arithmetic authority.
- **Adoption:** pilot team's monthly close and quarterly review use CostPilot outputs as the working papers, not parallel spreadsheets.
- **Value proof:** at least one ABC-driven pricing or product-mix decision taken at the pilot that quantifiably changes margin, and one reforecast completed in under one-third of the previous cycle time.

7. Key Risks and Mitigations

Risk	Mitigation
ABC implementation fatigue — the reason ABC adoption stalled industry-wide since the 1990s	TDABC as the default modeling mode; copilot maintains the model continuously so effort amortises; pilot selects a plant with available setup/inspection logs
ERP connector projects balloon into multi-year integrations	Read-only scope, one connector first, data-contract approvals gate any extension
Budget workflow features turn the product into an enterprise-planning suite (scope sprawl against dedicated EPM vendors)	Phase 2 budget scope limited to template chain + consolidation + scenarios; no form-based budget-capture UI build-out
Performance-measurement features trigger internal politics (who is judged on what)	Ship evaluation analytics as comparison views, not verdicts; controllability tagging owned by the client
Data quality at ERP scale exceeds pilot-grade assumptions	Reconciliation gates per ingestion run; period snapshots preserve the same-question-same-number guarantee

8. Looking Beyond: Phase 3 Preview

Out of scope here, recorded for continuity: marketing-management decision analytics (channel marginal ROI, CAC/LTV, campaign P&L, pricing elasticity and promotion economics, marketing-budget variance analysis), full intercompany automation, write-back workflows, and multi-industry cost templates. The marketing knowledge layer does not exist yet in our curriculum base and will need its own source canon before engine modules are specified against it.